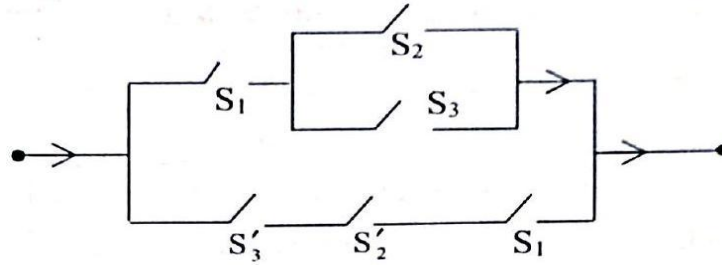


MATHEMATICS

- 101.** Write the statement in symbolic form 'Sandeep neither likes tea nor coffee but enjoys a Soft drink.' Where p : Sandeep likes tea, q : Sandeep likes coffee, r : Sandeep enjoys a soft Drink
- (A) $(\sim p \vee \sim q) \wedge r$ (B) $(\sim p \vee \sim q) \vee r$ (C) $(\sim p \wedge q) \vee r$ (D) $(\sim p \wedge \sim q) \wedge r$
- 102.** If the statement $p \vee \sim (q \wedge r)$ is false, then the truth values of p, q and r are respectively
- (A) F, T, F (B) T, F, F (C) F, T, T (D) F, F, T
- 103.** The contrapositive of the statement $\sim p \vee (q \wedge \sim r)$ is
- (A) $p \rightarrow q \wedge r$ (B) $(q \wedge r) \rightarrow p$ (C) $\sim q \vee \sim r \rightarrow p$ (D) $(r \vee \sim q) \rightarrow \sim p$
- 104.** S_1 : If -7 is an integer, then $\sqrt{-7}$ is a complex number
 S_2 : -7 is not an integer or $\sqrt{-7}$ is a complex number
- (A) S_1 and S_2 are converse statements of each other
 (B) S_1 and S_2 are negations of each other
 (C) S_1 and S_2 are equivalent statements
 (D) S_1 and S_2 are contrapositive of each other
- 105.** The statement $\sim (p \leftrightarrow \sim q)$
- (A) equivalent to $p \leftrightarrow q$ (B) a fallacy (C) a tautology (D) equivalent to $\sim p \leftrightarrow q$
- 106.** If q is false and $p \wedge q \leftrightarrow r$ is true, then ____ is a tautology.
- (A) $p \vee r$ (B) $(p \wedge r) \rightarrow p \vee r$ (C) $(p \vee r) \rightarrow p \wedge r$ (D) $p \wedge r$
- 107.** The statement pattern $[(p \rightarrow q) \wedge \sim q] \rightarrow r$ is a tautology when r is equivalent to
- (A) $p \wedge \sim q$ (B) $q \vee p$ (C) $p \wedge q$ (D) $\sim q$
- 108.** If $A = \{1, 2, 3, 4, 5\}$ then which of the following statement is not true?
- (A) $\exists x \in A$, such that $x + 4 \leq 9$ (B) $\exists x \in A$, such that $12 - x = 7$
 (C) $\forall x \in A, x + 3 \geq 4$ (D) $\exists x \in A$, such that $x + 2 < 3$
- 109.** Which of the following is the negation of the statement "For all $M > 0$, there exist $x \in s$ Such that $x \geq M$ "
- (A) $\exists M > 0$ such that $x \geq M$ for all $x \in s$ (B) $\exists M > 0, \exists x \in s$ such that $x \geq M$
 (C) $\exists M > 0$ such that $x < M$ for all $x \in s$ (D) $\exists M > 0$, there exist $x \in s$ such that $x < M$

110. If $p \equiv$ The switch S_1 is closed, $q \equiv$ the switch S_2 is closed, $r \equiv$ switch S_3 is closed, then symbolic form of following switching circuit is equivalent to



- (A) p (B) q (C) $p \wedge q$ (D) $p \vee q$

111. If $A^{-1} = \frac{-1}{2} \begin{bmatrix} 1 & -4 \\ -1 & 2 \end{bmatrix}$, then $2A + I_2 =$

- (A) $\begin{bmatrix} 5 & 8 \\ 1 & 2 \end{bmatrix}$ (B) $\begin{bmatrix} 5 & 8 \\ 2 & 2 \end{bmatrix}$ (C) $\begin{bmatrix} 2 & 4 \\ 1 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} 5 & 8 \\ 2 & 3 \end{bmatrix}$

112. For 3×3 matrix A ,

if $A(\text{adj}A) = \begin{bmatrix} -10 & 0 & 0 \\ 0 & -10 & 2 \\ 0 & 0 & -10 \end{bmatrix}$, then the value of determinant of A is

- (A) 100 (B) -10 (C) -1000 (D) 20

113. Given $A = \begin{bmatrix} x & 3 & 2 \\ 1 & y & 4 \\ 2 & 2 & z \end{bmatrix}$, if $xyz = 60$ and $8x + 4y + 3z = 20$, then $A(\text{adj}A)$ is equal to

- (A) $\begin{bmatrix} 60 & 0 & 0 \\ 0 & 60 & 0 \\ 0 & 0 & 60 \end{bmatrix}$ (B) $\begin{bmatrix} 108 & 0 & 0 \\ 0 & 108 & 0 \\ 0 & 0 & 108 \end{bmatrix}$ (C) $\begin{bmatrix} 20 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 20 \end{bmatrix}$ (D) $\begin{bmatrix} 68 & 0 & 0 \\ 0 & 68 & 0 \\ 0 & 0 & 68 \end{bmatrix}$

114. If $A + B = \begin{bmatrix} 1 & \tan \frac{\theta}{2} \\ -\tan \frac{\theta}{2} & 1 \end{bmatrix}$ where A is symmetric and B is skew-symmetric matrix, then

The matrix $(A^{-1}B + AB^{-1})$ at $\theta = \frac{\pi}{6}$ is given by

(A) $\begin{bmatrix} 1 & 2\sqrt{3} \\ 2\sqrt{3} & 1 \end{bmatrix}$ (B) $\begin{bmatrix} -1 & -2\sqrt{3} \\ 2\sqrt{3} & 1 \end{bmatrix}$ (C) $\begin{bmatrix} 0 & 2\sqrt{3} \\ 2\sqrt{3} & 0 \end{bmatrix}$ (D) $\begin{bmatrix} 0 & -2\sqrt{3} \\ 2\sqrt{3} & 0 \end{bmatrix}$

115. Let A be a non-singular matrix of order n and $|A| = k$, then $(\text{adj}A)^{-1}$ is

(A) $\frac{A}{k}$ (B) $k^{n-1}(\text{adj} A)$ (C) $k^{n-2} A$ (D) kA

116. If $\begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 4 \\ 1 & 3 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 12 \\ 15 \\ 13 \end{bmatrix}$, then the value of

$x^2 + y^2 + z^2 =$

(A) 6 (B) 12 (C) 3 (D) 14

117. If $A = \begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$, then $A^T \cdot A^{-1} =$

(A) $\begin{bmatrix} -\cos 2x & \sin 2x \\ -\sin 2x & \cos 2x \end{bmatrix}$ (B) $\begin{bmatrix} \cos 2x & -\sin 2x \\ \sin 2x & \cos 2x \end{bmatrix}$

(C) $\begin{bmatrix} \cos 2x & \sin 2x \\ -\sin 2x & \cos 2x \end{bmatrix}$ (D) $\begin{bmatrix} \cos 2x & -\sin 2x \\ -\sin 2x & \cos 2x \end{bmatrix}$

118. If A and B are square matrices of order 3 such that $|A| = 5$, $|B| = 2$, then $|A(\text{adj}B)| =$

(A) 20 (B) 64 (C) 10 (D) 7

119. The general solution of the equation $\tan^2 x = 1$ is

(A) $n\pi + \frac{\pi}{4}$ (B) $n\pi - \frac{\pi}{4}$ (C) $n\pi \pm \frac{\pi}{4}$ (D) $2n\pi \pm \frac{\pi}{4}$

120. The number of principal solutions of $\tan 2\theta = 1$ is

(A) One (B) Two (C) Three (D) Four

121. The general solution of the equation $\cot 4x = -1$ is

(A) $\frac{n\pi}{4} + \frac{4\pi}{15}, n \in Z$ (B) $\frac{n\pi}{4} + \frac{13\pi}{7}, n \in Z$ (C) $\frac{n\pi}{4} + \frac{11\pi}{9}, n \in Z$ (D) $\frac{n\pi}{4} + \frac{3\pi}{16}, n \in Z$

122. The number of solutions of $\sin^2 \theta = \frac{1}{2}$ in $[0, \pi]$ is

(A) two (B) three (C) four (D) one

123. The number of solutions of the equation $\tan^3 x - 3\tan x = 0$ in the interval $(0, 2\pi)$ is

(A) 1 (B) 7 (C) 5 (D) 3

124. The general solution of $\tan 3x = 1$ is

(A) $x = n\pi \pm \frac{\pi}{4}, n \in Z$ (B) $x = n\pi, n \in Z$

(C) $x = n\left(\frac{\pi}{3}\right) + \frac{\pi}{12}, n \in Z$ (D) $x = n\pi + \frac{\pi}{4}, n \in Z$

125. The principal solutions of the equation $\sqrt{3}\operatorname{cosec}x + 2 = 0$ are

- (A) $\frac{4\pi}{3}, \frac{5\pi}{3}$ (B) $\frac{2\pi}{3}, \frac{4\pi}{3}$ (C) $\frac{2\pi}{3}, \frac{5\pi}{3}$ (D) $\frac{\pi}{3}, \frac{2\pi}{3}$

126. The general solution of $\frac{1-\cos 2x}{1+\cos 2x} = 3$ is

- (A) $x = n\pi \pm \frac{\pi}{6}, n \in \mathbb{Z}$ (B) $x = 2n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$
(C) $x = 2n\pi \pm \frac{\pi}{6}, n \in \mathbb{Z}$ (D) $x = n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$

127. The general solution of $\cos 2x = \frac{1}{2}$ is

- (A) $2n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$ (B) $n\pi \pm \frac{\pi}{6}, n \in \mathbb{Z}$
(C) $2n\pi \pm \frac{\pi}{6}, n \in \mathbb{Z}$ (D) $n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$

128. The principal solutions of the equation $\sin\theta = \frac{1}{\sqrt{2}}$.

- (A) $\frac{\pi}{4}, \frac{3\pi}{4}$ (B) $\frac{2\pi}{3}, \frac{4\pi}{3}$ (C) $\frac{2\pi}{3}, \frac{5\pi}{3}$ (D) $\frac{\pi}{3}, \frac{2\pi}{3}$

129. If $y = \log_{\sin x} \tan x$, then $\left(\frac{dy}{dx}\right)_{x=\frac{\pi}{4}}$ has the value

- (A) $\frac{4}{\log 2}$ (B) $-3\log 2$ (C) $\frac{-4}{\log 2}$ (D) $3\log 2$

130. If $8f(x) + 6f\left(\frac{1}{x}\right) = x + 5$ and $y = x^2f(x)$, then $\frac{dy}{dx}$ at $x = -1$ is

- (A) 14 (B) -14 (C) $\frac{1}{14}$ (D) $-\frac{1}{14}$

131. If $y = \log_3(\log_3 x)$ then $\frac{dy}{dx}$ at $x = 3$ is

- (A) $\frac{1}{3}(\log 3)^{-3}$ (B) $\frac{1}{3}(\log 3)$ (C) $\frac{1}{3}(\log 3)^{-3}$ (D) $\frac{1}{3}(\log 3)^{-2}$

132. $\frac{d}{dx} \left[\sec \left(\cos^{-1} \left(\frac{6}{x} \right) \right) \right] =$

- (A) $\frac{1}{8}$ (B) 8 (C) $\frac{1}{6}$ (D) none of these

133. If $y = e^{\sin(\operatorname{cosec}^{-1}x)}$, then $\frac{dy}{dx} =$

- (A) $e^{\cos(\operatorname{cosec}^{-1}x)}$ (B) 0 (C) $\frac{e^x}{x^2}$ (D) $-\frac{e^x}{x^2}$

134. $\frac{d}{dx} \left(\cos^{-1} \left(\frac{x-\frac{1}{x}}{x+\frac{1}{x}} \right) \right) =$

- (A) $\frac{x^2+1}{x^2-1}$ (B) $\frac{2}{1+x^2}$ (C) $\frac{-1}{1+x^2}$ (D) $\frac{-2}{1+x^2}$

135. $y = \frac{\sqrt[3]{1+3x}\sqrt[4]{1+4x}\sqrt[5]{1+5x}}{\sqrt[7]{1+7x}\sqrt[8]{1+8x}}$. Then $\frac{dy}{dx}$ at $x = 0$ is

- (A) 3 (B) -1 (C) 1 (D) 2

136. If $x^2y^2 = \sin^{-1} \sqrt{x^2 + y^2} + \cos^{-1} \sqrt{x^2 + y^2}$, then $\frac{dy}{dx} =$

- (A) $\frac{-y}{x}$ (B) $\frac{y}{x}$ (C) $\frac{x}{y}$ (D) $\frac{-x}{y}$

137. The derivative of $f(\sec x)$ with respect to $g(\tan x)$ at $x = \frac{\pi}{4}$, where $f'(\sqrt{2}) = 4$

And $g'(1) = 2$, is

- (A) 2 (B) $\frac{1}{\sqrt{2}}$ (C) $\sqrt{2}$ (D) $\frac{1}{2\sqrt{2}}$

138. If $x = \log t$, and $y + 1 = \frac{1}{t}$, then $e^{-x} \frac{d^2x}{dy^2} + \frac{dx}{dy} =$

- (A) 2 (B) -1 (C) 0 (D) 1

139. All the points on the curve $y^2 = 4a \left[x + a \sin \left(\frac{x}{a} \right) \right]$, where the tangent is parallel to the X

Axis lie on

- (A) circle (B) parabola (C) straight line (D) None of these

140. A particle is moving on a straight line. The distance S travelled in time t is given by $S = at^2 + bt + 6$. If the particle comes to rest after 4 seconds at a distance of 16 m .from the starting point, then the acceleration of the particle is

- (A) $\frac{-3}{4}$ m/sec² (B) $\frac{-1}{2}$ m/sec² (C) -1 m/sec² (D) $\frac{-5}{4}$ m/sec²

141. A spherical iron ball 10 cm in radius is coated with a layer of ice of uniform thickness that melts at a rate of 50 cm³/min. When the thickness of ice is 5 cm, then the rate at which the thickness of ice decreases, is

- (A) $\frac{1}{54\pi}$ cm/min (B) $\frac{1}{36\pi}$ cm/min (C) $\frac{1}{18\pi}$ cm/min (D) $\frac{5}{6\pi}$ cm/min

142. A square plate is contracting at the uniform rate 3 cm²/sec, then the rate at which the

Perimeter is decreasing, when the side of the square is 15 cm, is

- (A) $\frac{1}{5}$ cm/sec (B) $\frac{2}{5}$ cm/sec (C) $\frac{1}{10}$ cm/sec (D) $\frac{3}{10}$ cm/sec

143. Using differentiation, the approximate value of $\sin 46^\circ$, given that $1^\circ = 0.0175^\circ$

- (A) 0.07194 (B) $\frac{1.0175}{\sqrt{2}}$ (C) 0.7194 (D) $\frac{0.0175}{\sqrt{2}}$

144. The approximate value of $\sqrt[3]{0.026}$ is

- (A) 0.2762 (B) 0.2963 (C) 0.2632 (D) 0.2692

145. Let $f(0) = -3$ and $f'(x) \leq 5$ for all real values of x . The $f(2)$ can have possible maximum value as

- (A) 10 (B) 5 (C) 7 (D) 13

146. The length of the longest interval, in which the function $3\sin x - 4\sin^3 x$ is increasing, is

- (A) $\frac{\pi}{3}$ (B) $\frac{\pi}{2}$ (C) $\frac{3\pi}{2}$ (D) π

147. If rectangles are inscribed in a circle of radius r units, then the dimensions of the rectangle

Which has maximum area are

- (A) $2r$ units, r units (B) $2r$ units, $\sqrt{2}r$ units,
(C) $\sqrt{2}r$ Units, $\sqrt{2}r$ units (D) r units, $\sqrt{2}r$ units

148. 20 meters of wire is available to fence of a flowerbed in the form of a circular sector. If the flowerbed is to have maximum surface area, then the radius of the circle is
(A) 2 m (B) 5 m (C) 4 m (D) 8 m

149. The minimum value of the slope of the tangent to curve $y = x^3 - 3x^2 + 2x + 93$ is
(A) 1 (B) -1 (C) 2 (D) -2

150. An open tank with a square bottom is to contain 4000 cubic cm. of liquid. The dimensions

Of the tank so that the surface area of the tank is minimum, is

- (A) side = 20 cm, height = 10 cm (B) side = 10 cm, height = 20 cm
(C) side = 10 cm, height = 40 cm (D) side = 20 cm, height = 05 cm